Why SAMURAI73 Needs Two Polarimeters

Full tensor reconstruction and locating p_{yy} loss

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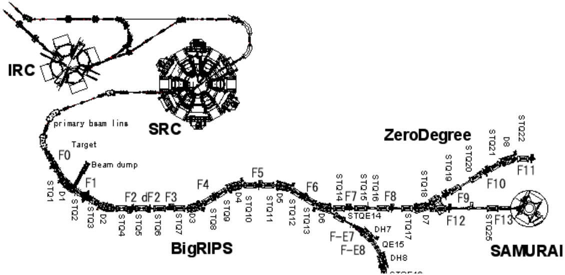
IVR needs the tensor state delivered to the target



First IVR run

Start with vertical p_{yy}; longitudinal p_{zz} is harder to prepare.

After SRC

What polarization leaves the accelerator chain?

After BigRIPS

What polarization reaches SAMURAI73?

Two stations are needed to determine all degrees of freedom of the spin-density matrix.

Spin-1: 8 general degrees, 5 tensor, 1 prepared

For $w_a \geq 0$ and $\sum_a w_a = 1$,

$$\rho = \sum_a w_a |\psi_a\rangle\langle\psi_a|.$$

A normalized 3×3 Hermitian ρ has

$$\boxed{8 = 3_{\text{vector}} + 5_{\text{tensor}}}$$

independent real parameters.

An axial tensor prepared about $\hat{\boldsymbol{n}}$ is

$$\boldsymbol{P}_0^{(2)}(q, \hat{\boldsymbol{n}}) = \frac{q}{2}\left(3\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^{\mathsf{T}} - I\right).$$

$$\hat{\boldsymbol{n}} = \hat{\boldsymbol{y}} : \quad \boldsymbol{P}_0^{(2)} = \frac{p_{yy}}{2}\,\mathrm{diag}(-1, 2, -1),$$

$$\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}} : \quad \boldsymbol{P}_0^{(2)} = \frac{p_{zz}}{2}\,\mathrm{diag}(-1, -1, 2).$$

Tensor only: 5 degrees. Fixed preparation axis: one amplitude q.

Linear BMT preserves vector and tensor ranks

1. Magnetic field rotates the spin in the lab

$$\rho_{\mathrm{lab},f} = U_s \rho_0 U_s^\dagger, \qquad R_s \in SO(3).$$

Thomas–BMT determines R_s along the trajectory.

2. The downstream beam defines new axes

$$\rho_{b,f} = U_o^\dagger \rho_{\mathrm{lab},f} U_o.$$

R_o follows the transported momentum and defines the local x_b, y_b, z_b frame.

$$\boxed{R_{\mathrm{rel}} = R_o^{\mathsf{T}} R_s}, \qquad \boldsymbol{p}_{b,f} = R_{\mathrm{rel}} \boldsymbol{p}_0, \qquad \boldsymbol{P}_{b,f}^{(2)} = R_{\mathrm{rel}} \boldsymbol{P}_0^{(2)} R_{\mathrm{rel}}^{\mathsf{T}}.$$

Lowest order: a common rotation moves the polarization axis, but vector and tensor ranks remain separate.

BMT field spread cannot convert tensor to vector

Each phase-space point $\xi = (x, x', y, y', \Delta p/p, \ldots)$ has its own calculated rotation:

$$\boldsymbol{p}_f(\xi) = R_\xi \boldsymbol{p}_0, \qquad \boldsymbol{P}_f^{(2)}(\xi) = R_\xi \boldsymbol{P}_0^{(2)} R_\xi^{\mathrm{T}}.$$

The polarimeter measures the ensemble average:

$$\overline{\boldsymbol{p}}_f = \int d\xi\, f(\xi) R_\xi \boldsymbol{p}_0, \qquad \overline{\boldsymbol{P}^{(2)}}_f = \int d\xi\, f(\xi) R_\xi \boldsymbol{P}_0^{(2)} R_\xi^{\mathrm{T}}.$$

If $\boldsymbol{p}_0 = 0$, then exactly

$$\overline{\boldsymbol{p}}_f = 0.$$

The averaged tensor can become biaxial and require all

$$\boxed{\text{5 tensor components}}.$$

Different $B(\xi)$ under linear BMT can open up to 5 tensor degrees; it cannot give tensor→vector conversion.

Rank mixing opens all 8 state parameters

Tensor–vector conversion requires spin dynamics beyond the linear BMT term. For example,

$$H_{\mathrm{spin}} = -\mu\, \boldsymbol{S}\cdot\boldsymbol{B} - \beta_T \left(\boldsymbol{S}\cdot\boldsymbol{B}\right)^2 + \cdots, \qquad \overline{\rho}_f = \int d\xi\, f(\xi)\, \mathcal{E}_\xi[\rho_0].$$

Quadratic spin interaction
Tensor magnetic/electric polarizability or other quadratic spin couplings can generate

$$\boldsymbol{p}_0 = 0,\ \boldsymbol{P}_0^{(2)} \neq 0 \quad \longrightarrow \quad \boldsymbol{p}_f \neq 0.$$

Spin-dependent loss or material interaction
Spin-dependent loss, material interactions, collimation, and acceptance can change both ranks and the density-matrix eigenvalues.

Once rank mixing is allowed, the fit must allow the full $8 = 3_{\mathrm{vector}} + 5_{\mathrm{tensor}}$ parameters.

The cross section shows the two blind directions

$$
\begin{aligned}
\frac{\sigma(\theta,\phi)}{\sigma_0(\theta)} = 1 &+ \frac{3}{2}\left(p_x \sin\phi + p_y \cos\phi\right) A_y(\theta) \\
&+ \frac{2}{3}\left(p_{xz} \cos\phi - p_{yz} \sin\phi\right) A_{xz}(\theta) \\
&+ \frac{1}{6}\left[(p_{xx} - p_{yy}) \cos 2\phi - 2p_{xy} \sin 2\phi\right]\left[A_{xx}(\theta) - A_{yy}(\theta)\right] \\
&+ \frac{1}{2} p_{zz} A_{zz}(\theta).
\end{aligned}
$$

p_z is absent

The d–p reaction has no longitudinal-vector term in this geometry.

p_{xy} vanishes at the four arms

At L/R/U/D, $\phi = 0^\circ, 90^\circ, 180^\circ, 270^\circ$, so $\sin 2\phi = 0$.

One calibrated multi-angle station measures 6/8 density-matrix parameters, or 4/5 tensor parameters.

BigRIPS makes both blind directions visible

Before BigRIPS: unprimed. **After BigRIPS:** primed.

$$R_{\mathrm{rel}} = R_y(\delta), \qquad \delta = 27.513^\circ, \qquad c = \cos\delta = 0.886906, \qquad s = \sin\delta = 0.461951.$$

Vector blind direction

$$p'_x = c\,p_x + s\,p_z,$$
$$p'_z = -s\,p_x + c\,p_z.$$

Hence an upstream p_z gives

$$\boxed{p'_x = 0.461951\,p_z},$$

which is measured at station 2.

Tensor blind direction

$$p'_{xy} = c\,p_{xy} + s\,p_{yz},$$
$$p'_{yz} = -s\,p_{xy} + c\,p_{yz}.$$

Hence an upstream p_{xy} gives

$$\boxed{p'_{yz} = -0.461951\,p_{xy}},$$

which is measured at station 2.

Two stations: tensor 5/5; full density matrix 8/8.

Why can p_{yy} decrease?

Compare tensor size Q_T, vector polarization $V = |\boldsymbol{p}|$, and density-matrix purity $\Pi = \mathrm{Tr}(\rho^2)$.

what the polarimeters see	what happened in the beam
p_{yy} **changes;** Q_T **and** Π **stay constant; the axis moves;** $V = 0$	Common magnetic rotation: all particles' axes turn together; no polarization is lost.
Q_T **changes;** V **appears;** Π **stays constant**	Quadratic spin coupling transfers polarization from tensor to vector.
Π **decreases**	Different trajectories acquire different axes or phases and cancel in the average; spin-dependent loss can also change ρ.

Changed already after SRC $\Rightarrow$ upstream. Changed only after BigRIPS $\Rightarrow$ between the stations.

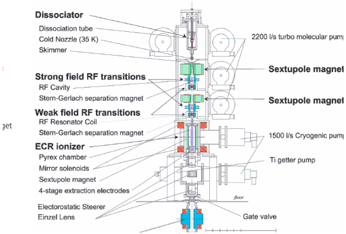
After-SRC installation: contact and procedure

1. Who should we contact at RIKEN?
2. What approval and installation procedure should we follow?

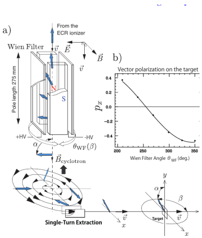
Backup

Source, accelerator, beam-line, and detector details

Backup: Wien filter sets the PIS spin axis



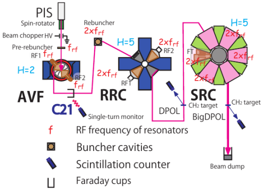
RIKEN polarized ion source



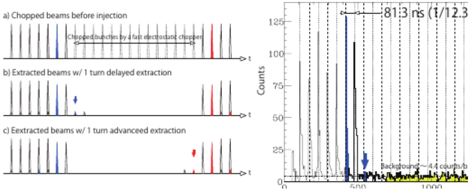
Wien-filter spin-orientation control

- The two-stage PIS produces vector and tensor modes.
- After the ECR ionizer the axis is vertical; crossed $\boldsymbol{E}$ and $\boldsymbol{B}$ tilt the spin without changing the orbit.
- Rotating the Wien filter sets the azimuth; vector polarization calibrates the angles.

Backup: single-turn extraction preserves spin phase



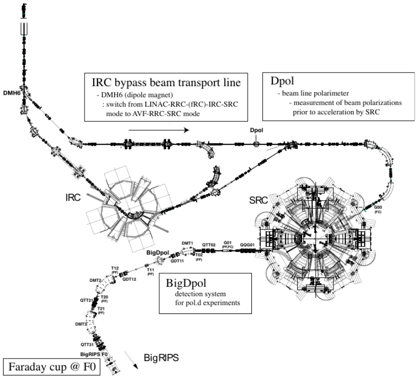
PIS → AVF → RRC → SRC

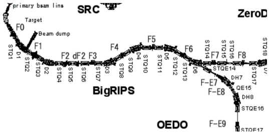


Timing signature of wrong-turn extraction

- The cyclotron field changes the spin direction every turn: $\Delta\beta = 360^\circ\gamma(g-1)$, with $g = 0.8477$ in the paper's convention.
- Mixing N with $N \pm 1$ turns superposes spin phases and lowers the polarization.
- AVF, RRC, and SRC kept the same turn: 0.07% mixing, $> 99\%$ purity, stable for 2–3 days.

Backup: magnets between the two stations





- Station 1: BigDpol/T11, between DMT1 and DMT2.
- DMT2 and DMT3: two identical 50° dipoles.
- BigRIPS $F0 \to F7$: six 30° dipoles, signs $(-,-,-,+,+,-)$, net bend $-60°$.
- $F7 \to F13$: quadrupoles, but no incident-beam dipole before the target.

Backup: central ray gives $\delta = 27.513^\circ$ at 190 MeV/u

DMT2 + DMT3	-100°
BigRIPS $F0 \to F7$	-60°
$F7 \to F13$	0°
total orbit bend θ	-160°

For a 190 MeV/u deuteron,

$$\gamma = 1.20260044, \qquad G_d = -0.14298727,$$

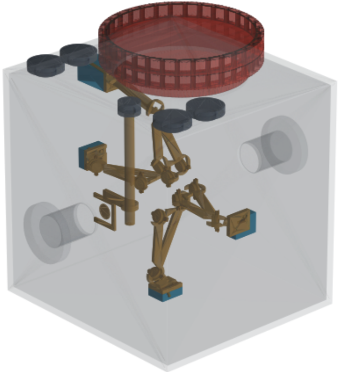
$$\delta = G_d\gamma\theta = +27.513^\circ, \qquad R_{\mathrm{rel}} = R_y(\delta).$$

The central rotation exposes the one-station blind directions:

$$p'_x = 0.461951\, p_z + \cdots\,, \qquad p'_{yz} = -0.461951\, p_{xy} + \cdots\,.$$

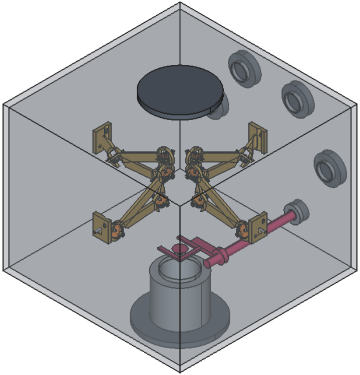
This is the central-ray horizontal-bend baseline; the real analysis uses $R_{\mathrm{rel}}(\xi)$ from field maps and measured beam phase space.

Backup: proposed polarimeter after SRC



Compact in-vacuum polarimeter proposed just downstream of SRC.

Backup: proposed polarimeter before SAMURAI



Compact in-vacuum polarimeter proposed upstream of SAMURAI.

Backup: expected p_{yy} precision and rates

Detector geometry

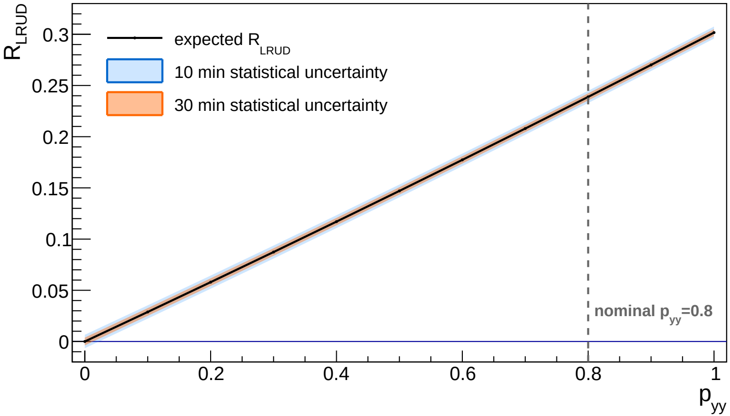
channel	θ_{lab}	radius
deuteron	20.9°	140 mm
proton, small angle	11.2°	190 mm
proton, large angle	53.4°	205 mm

Rate assumptions

beam	190 MeV/u, 10^7 d/s
nominal polarization	$p_{yy} = 0.8$
CH_2 for rate model	$1000\,\mathrm{mg/cm^2}$
selected LR/UD rate	$36.95\,\mathrm{s^{-1}}$

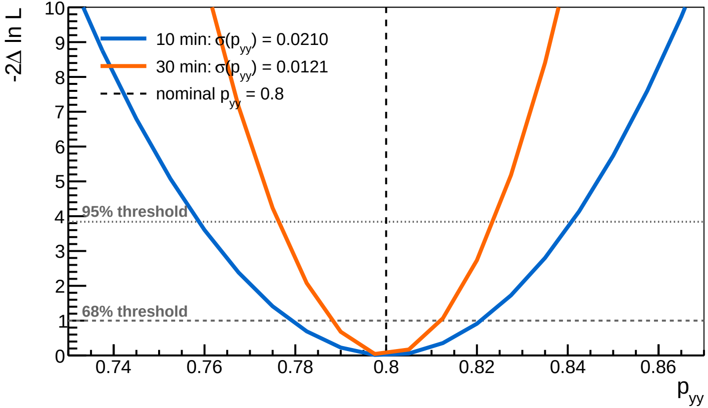
time	$N_L + N_R$	$N_U + N_D$	total	p_{yy} statistical precision
10 min	13,734	8,438	22,172	$\sigma = 0.0210$; 68% CI $[0.779, 0.821]$
30 min	41,203	25,314	66,517	$\sigma = 0.0121$; 68% CI $[0.788, 0.812]$

Backup: LR/UD count ratio versus p_{yy}



At $p_{yy} = 0.8$**:** $R_{LRUD} = 0.2389$**;** $\sigma_R = 0.0060$ **(10 min) and** 0.0035 **(30 min).**

Backup: p_{yy} precision at 10 and 30 min



At $p_{yy} = 0.8$: $\sigma = 0.0210$ at 10 min and 0.0121 at 30 min (42% smaller).